

EPFO website remake

Selected workflows, pain-point mapping, and system design for a demo built for everyone

Four workflows were selected for the demo based on impact and relatability: UAN Activation, View Passbook / Balance Check, Job Switch / Transfer, and Online Withdrawal / Claim. Together they form a natural lifecycle story — new user, daily user, job change, moment of need.

Pain-point to solution mapping

Workflow	Pain points	UX fix (non-tech-savvy first)	System design fix (scale)
UAN Activation	Face-auth fails on low-end phones / poor lighting; no regional-language help; crashes during mass hiring drives	One-screen guided flow; voice instructions in regional languages; WhatsApp/IVR fallback for feature phones; a short “why this matters” explainer before the Aadhaar step	Queue-based throttling to the Aadhaar API with retry/backoff; horizontally-scaled activation service; idempotent requests so retries are safe; pre-warmed capacity for known hiring seasons
View Passbook / Balance	Slow loads (live pull across employers); jargon-heavy line items; no offline access; missed-call/SMS balance often fails silently	Plain-language split of “your money” / “employer’s money” / “interest earned”; downloadable PDF; low-data mode; a visible “last updated” timestamp	Precomputed, cached passbook summaries (read replicas, CQRS); ECR ingestion fully decoupled from reads; database sharded by UAN hash; CDN-cached summaries
Job Switch / Transfer	Zero visibility after submission (“did my money follow me?”); employer delays in linking; silent failures	Visual status tracker (Submitted → Employer verified → Merged); automatic SMS/push at every stage; no manual “transfer request” step, matching the real auto-merge design	Event-driven state machine per transfer instance; employer portal emits events asynchronously instead of blocking; idempotent merge logic to prevent duplicate or lost balances; pub/sub notifications
Withdrawal / Claim	Highest emotional stakes (job loss, medical, marriage); silent KYC-mismatch rejections discovered weeks later; legal-language forms; no live status; long queues during layoffs	Pre-submission eligibility checker that flags KYC mismatches before submission; plain-language rejection reasons; live status tracker; instant document validation; save-and-resume; multilingual	Pre-validation service filters bad requests before the core claims pipeline; async workflow orchestration; priority queue for urgent cases (job loss/medical); autoscaling for layoff-driven spikes; circuit breakers on bank-verification calls

System architecture (built to hold lakhs of concurrent requests)

Client apps (web, WhatsApp, IVR/SMS) → API gateway (rate limiting, auth, edge cache) → four independent workflow microservices (Activation, Passbook, Transfer, Claims) → an async event queue that absorbs traffic spikes → a sharded data layer with read replicas. The Status & Grievance service sits alongside the gateway and reads from the same monitoring layer as everything else.

Cross-cutting scale principles

- Read-heavy traffic (passbook checks) is served from cache and read replicas, fully decoupled from write-heavy traffic (claims, transfers), which flows through the queue to absorb bursts.
- Each workflow is its own microservice, so one overloaded flow cannot take the others down.
- Autoscaling is tied to known peak periods: fiscal year-end, mass layoffs, hiring season.
- Circuit breakers and retry/backoff wrap every external dependency (Aadhaar, bank verification APIs).
- All system health metrics feed into the Status page in real time.

Status & grievance page (separate, standalone)

- Live green / yellow / red status indicators per workflow, plus a “known issues” banner for transparency.
- Grievance registration and tracking by ticket number — no login required, so it works for anyone.
- Search by tracking ID, with SMS/push updates whenever status changes.